

Glossray

Table of Contents

Chapter 1:	2
Chapter 2:	5
Chapter 3:	6
Chapter 4:	8
Chapter 5:	9
chapter 6:	10
Chapter 7:	11
Chapter 8:	12
Chpater 14:	13

Chapter 1:

1. **Business Statistics:** The collection of procedures and techniques used to convert data into meaningful information in a business environment.
2. **Data:** Raw facts, figures, or information collected for analysis.
3. **Variables:** Characteristics or attributes that can vary and be measured.
4. **Descriptive Statistics:** Procedures and techniques designed to summarise and describe data.
5. **Inferential Statistics:** Tools and techniques used to draw conclusions about a population based on a sample.
6. **Estimation:** The process of estimating population parameters based on sample data.
7. **Hypothesis Testing:** A method of using sample data to make inferences about a hypothesis related to population characteristics.
8. **Experiment:** A process that produces a single outcome whose result cannot be predicted with certainty.
9. **Experimental Design:** A plan for performing an experiment in which the variable of interest is defined.
10. **Closed-End Questions:** Questions that require respondents to select from a defined list of choices.
11. **Demographic Questions:** Questions relating to respondents' characteristics, backgrounds, and attributes.
12. **Open-End Questions:** Questions that allow respondents freedom to respond with their own words or statements.
13. **Direct Observations:** Data collection through physically observing and recording what takes place.
14. **Personal Interviews:** Data collection involving structured or unstructured questioning of individuals.
15. **Structured Interviews:** Interviews using a specific set of predetermined questions, asked in the same order to all respondents, with no follow-up questions.

16. **Unstructured Interviews:** Interviews using no predetermined set of questions, beginning with a general question and allowing the discussion to flow, with probes for clarification.
17. **Focus Groups:** A group interview using a facilitator to obtain group perspectives and opinions.
18. **Documents and Records:** Data collection by studying existing secondary data in databases, reports, etc.
19. **Bar Codes:** Printed code that provides information about a product, which can be scanned to retrieve data.
20. **RFID (Radio Frequency Identification Device):** A transmitting device that attaches to a product to transmit data to a receiver.
21. **Interviewer Bias:** Bias introduced by the interviewer, affecting participant responses.
22. **Nonresponsive Bias:** Bias resulting from certain individuals not responding to the survey.
23. **Selection Bias:** Bias caused by a non-random selection of participants for the study.
24. **Observer Bias:** Bias introduced by the observer recording or interpreting data inaccurately.
25. **Measurement Error:** Errors in the data resulting from inaccurate measurements.
26. **Internal Validity:** The degree to which the research is actually measuring what it intends to.
27. **External Validity:** The degree to which the research results can be generalised to other contexts and populations.
28. **Data Accuracy:** The degree to which data is free from errors.
29. **Population:** The entire set of objects or individuals of interest.
30. **Sample:** A subset of the population.
31. **Census:** An enumeration of the entire population.
32. **Parameters:** Descriptive numerical measures computed from an entire population.
33. **Statistics:** Descriptive numerical measures computed from a sample.

34. **Statistical Sampling:** Sampling methods that use techniques based on chance selection.
35. **Nonstatistical Sampling:** Sampling methods using convenience, judgment, or other non-chance processes.
36. **Convenience Sampling:** A non-statistical sampling method that selects items in the most convenient manner.
37. **Judgment Sampling:** A non-statistical sampling method based on judgments about who would provide the most relevant information.
38. **Ratio Sampling:** A non-statistical sampling method where the sample size selected from a population segment is proportional to the segment's size.
39. **Simple Random Sampling:** Every possible sample has an equal chance of being selected.
40. **Stratified Random Sampling:** The population is divided into subgroups (strata), and a simple random sample is taken from each subgroup.
41. **Systematic Random Sampling:** Sampling method in which every kth item is selected after randomly selecting a start point within the population.
42. **Cluster Random Sampling:** Population is divided into clusters, a random sample of clusters is selected, and all or some items within those clusters are included in the sample.
43. **Quantitative Data:** Measurements with inherently numerical values (discrete or continuous).
44. **Qualitative Data:** Data with inherently categorical measurement scales.
45. **Time-Series Data:** Consecutive data values observed at successive points in time.
46. **Cross-Sectional Data:** A set of data values observed at a fixed point in time.
47. **Nominal Data:** Data classified into categories without any inherent order or ranking.
48. **Ordinal Data:** Data that can be ranked, but the intervals between the ranks do not need to be equal or measurable.
49. **Interval Data:** Data where the intervals between values are equal and measurable, but there is no true zero point.
50. **Ratio Data:** Data with a true zero point, allowing for ratios between values to be calculated.

Chapter 2:

1. **Bar Chart:** A graphical representation of categorical data using rectangles (bars) where the length or height of each bar represents the frequency, percentage, or another measure associated with the category.
2. **Class Boundaries:** The upper and lower values defining each class in a grouped data frequency distribution.
3. **Cumulative Frequency Distribution:** A summary of data displaying the number of observations with values less than or equal to the upper limit of each class.
4. **Cumulative Relative Frequency Distribution:** A summary of data displaying the proportion of observations with values less than or equal to the upper limit of each class.
5. **Continuous Data:** Data that can take any value within a given range; also known as uncountable data. For example, weight, length, and time.
6. **Dependent Variable:** In a scatter diagram, this is the variable whose values are thought to be a function of, or influenced by, the other variable.
7. **Discrete Data:** Data that can only take on a countable number of possible values; for example, student ages, number of product categories purchased.
8. **Frequency Distribution:** A summary of a data set showing the number of observations within each distinct category or class.
9. **Frequency Histogram:** A graph displaying a frequency distribution, where the horizontal axis contains possible outcomes and the vertical axis shows frequencies, represented by bars with no gaps between them.
10. **Grouped Data Frequency Distribution:** A summary of data where data is organized into classes, usually used with continuous data or discrete data with many possible outcomes.
11. **Independent Variable:** In a scatter diagram, this is the variable whose values are thought to impact the values of the dependent variable.
12. **Joint Frequency Distribution:** A table showing the frequencies of the occurrences of combinations of two or more variables.
13. **Line Chart:** A two-dimensional chart showing time on the horizontal axis and the variable of interest on the vertical axis. It is used to display time-series data.

14. **Pareto Chart:** A specific type of bar chart used to separate the "vital few" from the "trivial many," with bars representing categorical values and organized from high to low.
15. **Pie Chart:** A circular graph divided into "slices" to represent proportions of a total, where the size of each slice corresponds to the magnitude of the displayed variable associated with each category.
16. **Relative Frequency:** The proportion of total observations within a given category.
17. **Scatter Diagram (Scatter Plot):** A two-dimensional graph of plotted points, where each axis represents a quantitative variable, used to show relationships between two variables.
18. **Stem and Leaf Diagram:** A diagram used to present quantitative data in a way that shows the distribution of values while retaining the original values.

Chapter 3:

1. **Parameter:** A measure computed using data from the entire population.
2. **Statistic:** A measure computed using data from a sample taken from a population.
3. **Mean:** The arithmetic average of a set of data, calculated by summing all values and dividing by the number of values. Also known as the average.
4. **Median:** The middle value in an ordered dataset, dividing the data into two equal halves.
5. **Mode:** The value that appears most frequently in a dataset.
6. **Weighted Mean:** The mean of a dataset where each value is given a certain weight, reflecting its relative importance.
7. **Data Array:** A list of data that has been arranged in numerical order.
8. **Interquartile Range (IQR):** The difference between the third quartile (Q3) and the first quartile (Q1) of a dataset; measures the spread of the middle 50% of the data.
9. **Range:** The difference between the maximum and minimum values in a dataset; a simple measure of dispersion.

10. **Variance:** The average of the squared differences between each data value and the mean; a measure of data dispersion in units squared.
11. **Standard Deviation:** The square root of the variance; a measure of the typical deviation from the mean, in the same units as the data.
12. **Coefficient of Variation (CV):** A measure of relative variability, calculated by dividing the standard deviation by the mean, expressed as a percentage; allows for comparison of dispersion across datasets with different means and/or units.
13. **Empirical Rule:** A statistical rule stating that, for bell-shaped distributions, approximately 68% of the data values fall within one standard deviation of the mean, 95% within two standard deviations, and virtually all values within three standard deviations.
14. **Z-score:** The number of standard deviations a data value is away from the mean. Also known as a standardised data value. It quantifies how unusual a data point is in the context of its distribution.
15. **Percentiles:** Values that divide a dataset into 100 equal parts. The p th percentile is a value below which p percent of the data falls.
16. **Quartiles:** Values that divide a dataset into four equal parts. The first quartile (Q1) is the 25th percentile, the second quartile (Q2) is the 50th percentile (median), and the third quartile (Q3) is the 75th percentile.
17. **Box-and-Whisker Plot:** A graphical representation of a dataset using a box to represent the IQR, a line for the median, and whiskers that extend to the data's limits, with outliers marked with asterisks.
18. **Outlier:** A data point that falls significantly outside the expected range of values in a dataset.
19. **Nominal Data:** Data that represents categories with no intrinsic numerical order or ranking.
20. **Ordinal Data:** Data that represents categories with a meaningful rank order, but with non-uniform differences between the ranks.
21. **Interval Data:** Data that is measured on a scale with uniform intervals, but has an arbitrary zero point.
22. **Ratio Data:** Data that is measured on a scale with uniform intervals, and has a meaningful zero point.

Chapter 4:

1. **Probability:** The chance that a particular event will occur, expressed as a value between 0 and 1.
2. **Experiment:** A process that produces a single outcome, the result of which cannot be predicted with certainty.
3. **Sample Space:** The collection of all possible outcomes that can result from a selection, decision, or experiment.
4. **Mutually Exclusive Events:** Events that cannot occur at the same time; the occurrence of one precludes the occurrence of the other.
5. **Independent Events:** Events where the occurrence of one does not influence the probability of the occurrence of the other.
6. **Dependent Events:** Events where the occurrence of one impacts the probability of the other occurring.
7. **Classical Probability:** A method of assigning probability based on the ratio of favorable outcomes to total equally likely outcomes.
8. **Relative Frequency:** A method of assigning probability based on the number of times an event occurs divided by the total number of trials.
9. **Subjective Probability:** A method of assigning probability based on a decision-maker's state of mind or belief regarding the chances that a particular event will occur.
10. **Addition Rule:** The method used to find the probability of the union of two events (either one or the other). In the general form this will be $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.
11. **Complement Rule:** The probability that an event will not occur (given as $1 - P(A)$).
12. **Conditional Probability:** The probability of an event occurring given that another event has already occurred (given as $P(A|B)$).
13. **Multiplication Rule:** The method used to find the probability of the intersection of two events (both occurring). In the general form this will be $P(A \text{ and } B) = P(A) * P(B|A)$

14. **Tree Diagram:** A visual tool used to define the sample space, particularly for experiments with multiple steps, showing all possible outcomes and their probabilities.
15. **Bayes' Theorem:** A mathematical formula that describes how to update the probability of a hypothesis based on new evidence.

Chapter 5:

1. **Discrete Random Variable:** A variable that can only assume a finite number of values or a countably infinite sequence of values (e.g., 0, 1, 2, ...).
2. **Continuous Random Variable:** A variable that can assume an uncountably infinite number of values within a given range.
3. **Expected Value ($E(x)$):** The mean of a discrete probability distribution, representing the average outcome when the experiment generating the discrete random variable is repeated over a long run. It is calculated as the sum of each outcome's value multiplied by its probability ($\sum xP(x)$).
4. **Standard Deviation:** A measure of the spread or dispersion of a set of data or the values of a random variable around the mean, calculated as the square root of the variance.
5. **Binomial Distribution:** A probability distribution that describes the number of successes in a fixed number of independent trials, where each trial has only two possible outcomes (success or failure).
6. **Trial:** A single instance or observation in an experiment.
7. **Success:** The desired outcome in a binomial trial.
8. **Failure:** The opposite of a success in a binomial trial.
9. **Probability of Success (p):** The likelihood of a success occurring in a single binomial trial.
10. **Probability of Failure (q):** The likelihood of a failure occurring in a single binomial trial; calculated as $1 - p$.
11. **n (Binomial):** The fixed number of trials in a binomial experiment.

12. **x (Binomial):** The number of successes in 'n' trials of a binomial experiment.
13. **Counting Rule for Combinations:** A mathematical method for determining the number of ways events can occur, especially useful in binomial distributions where order does not matter.
14. **Cumulative Binomial Table:** A tool that gives the cumulative probability of obtaining a certain number or fewer of successes in a binomial distribution.
15. **Poisson Distribution:** A probability distribution that describes the number of events occurring in a fixed interval of time or space, particularly when these events happen at random.
16. **λ (lambda - Poisson):** The average number of events occurring in a given interval or segment in a Poisson distribution.
17. **t (Poisson):** The number of segments or intervals being considered in a Poisson distribution.
18. **λt (Poisson):** The average number of events occurring in 't' segments or intervals, calculated as λ multiplied by t.

chapter 6:

1. **Continuous Random Variable:** A variable that can take any value within a given range.
2. **Normal Distribution:** A bell-shaped, symmetrical probability distribution characterized by its mean (μ) and standard deviation (σ).
3. **Unimodal:** Having one mode or peak.
4. **Symmetrical Distribution:** A distribution in which the two halves of the curve are mirror images of each other.
5. **Mean (μ):** The average value of a dataset. In a normal distribution, it represents the centre of the distribution.
6. **Median:** The middle value in a dataset when ordered from smallest to largest.
7. **Mode:** The most frequently occurring value in a dataset.
8. **Standard Deviation (σ):** A measure of the spread or dispersion of data around the mean.

9. **Probability Density Function:** A function that describes the probability of a continuous random variable falling within a certain range.
10. **Standard Normal Distribution:** A normal distribution with a mean of 0 and a standard deviation of 1.
11. **z-value:** A measure of how many standard deviations a data point is from the mean in a standard normal distribution.
12. **Cumulative Probability:** The probability of a random variable taking on a value less than or equal to a given value.

Chapter 7:

1. **Central Limit Theorem:** A fundamental theorem in statistics stating that the distribution of sample means will approach a normal distribution as the sample size increases, regardless of the shape of the population distribution.
2. **Parameter:** A numerical value that describes a characteristic of a population (e.g., the population mean, μ , or the population standard deviation, σ).
3. **Population Proportion (p):** The fraction of values in a population that have a specific attribute of interest. It is calculated as X / N , where X is the number of items with the attribute, and N is the total population size.
4. **Sample Mean ($\bar{x}$):** The average of the values in a sample, used to estimate the population mean.
5. **Sample Proportion ($\hat{p}$):** The fraction of items in a sample that have a specific attribute of interest. It is calculated as the number of items with the attribute divided by the sample size.
6. **Sampling Distribution:** The distribution of a statistic (e.g. sample means or proportions) calculated from many different samples of the same size, taken from the same population.
7. **Sampling Error:** The difference between a sample statistic (like a sample mean or proportion) and the corresponding population parameter.

8. **Statistic:** A numerical value that describes a characteristic of a sample (e.g., sample mean, $\bar{x}$, or sample standard deviation, s).
9. **Z-Value:** A standardised score that measures the number of standard deviations a particular value is from the mean of its distribution.

Chapter 8:

1. **Point Estimate:** A single value derived from a sample that is used to approximate a population parameter (e.g., the sample mean as an estimate of the population mean).
2. **Confidence Interval Estimate:** An interval or range, constructed from a sample of data, used to estimate a population parameter with a specific level of confidence.
3. **Confidence Level:** The probability that a confidence interval, if repeatedly constructed, would contain the true population parameter. Usually expressed as a percentage (e.g., 95%).
4. **Critical Value:** The point on a probability distribution (normal or t) that corresponds to the desired confidence level, used in calculating the margin of error.
5. **Margin of Error:** The maximum likely difference between the point estimate and the true population parameter, equal to half of the width of the confidence interval.
6. **Population Parameter:** A numerical value that describes a characteristic of an entire population (e.g., the population mean or proportion).
7. **Sampling Error:** The difference between a sample statistic and the corresponding population parameter, arising from the randomness of sampling.
8. **Standard Normal Distribution:** A normal distribution with a mean of 0 and a standard deviation of 1, often used when the population standard deviation is known or for very large sample sizes.
9. **t-distribution:** A probability distribution similar to the normal distribution, but with heavier tails, used when the population standard deviation is unknown and the sample size is small.

Chapter 14:

1. **Bivariate Normal Distribution:** A joint probability distribution for two variables, where each variable is normally distributed and their scatter plot takes an elliptical shape.
2. **Coefficient of Determination (R^2):** The proportion of variance in the dependent variable that is predictable from the independent variable(s), ranging from 0 to 1. It shows the 'goodness of fit' of the model.
3. **Correlation Coefficient (r):** A measure of the strength and direction of the linear relationship between two variables. Ranges from -1 to +1.
4. **Dependent Variable:** A variable whose value is thought to depend on or be influenced by another variable (the independent variable); the 'y' variable.
5. **Independent Variable:** A variable thought to influence or explain the variance in the dependent variable; also known as an explanatory variable; the 'x' variable.
6. **Least Squares Criterion:** A method used to find the best-fitting line in regression analysis by minimizing the sum of the squares of the differences (residuals) between the actual and predicted values of the dependent variable.
7. **Linear Regression:** A statistical technique used to model the linear relationship between a dependent variable and one or more independent variables. Simple regression involves just one independent variable.
8. **Null Hypothesis:** A statement of "no effect" or "no difference" that is tested against an alternative hypothesis during statistical analysis.
9. **Residual:** The difference between the actual observed value of the dependent variable and the predicted value according to the regression model.
10. **Scatter Plot:** A two-dimensional plot showing the values for the joint occurrence of two quantitative variables. It graphically represents the relationship between the variables.
11. **Standard Error of the Slope:** A measure of the variability in the estimated slope of the regression line, used in hypothesis testing.
12. **Sum of Squares Error (SSE):** A measure of the unexplained variation around the regression line.

13. **Sum of Squares Regression (SSR):** A measure of the variation in the dependent variable that is explained by the independent variable(s).
14. **Total Sum of Squares (SST):** A measure of the total variation in the dependent variable.